Supplementary evidence: beta reduction and physical clearance

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This supplement preserves the original main-branch campaign. It distinguishes historical optical statistics from the new particle/clearance comparison. Original reviewed assets and campaign records remain frozen; historical claims are interpreted with their original beam and modelling limitations.

1 Original matching and local response

The historical three-start replay reproduces one feasible success, one infeasible success and one budget failure (Table 1). The successful infeasible candidate illustrates why matching and feasibility are separate; the failed candidate is not another low-residual match. Three starts do not establish optimizer reliability, search topology or feasible volume.

Table 1: Historical replay outcomes. The initial start is nominal, not random. J is the scaled residual sum of squares. Feasibility is assessed after search.

Start	J	peak β_x [m]	peak β_y [m]	outcome
Nominal	1.29×10^{-13}	31.437	50.746	feasible success
Random 1	1.21×10^{-12}	25.329	62.556	infeasible success
Random 2	1284.86	69.921	65.043	budget failure

The archived raw/rounded strengths remain feasible at 40-step resolution and independent-map precision. Table 2 supplies numerical values; decimal digits do not prescribe power-supply accuracy. Optical functions, dispersion and projected RMS sizes appear in Fig. 1. Beta peaks reported here are element-boundary samples rather than continuous maxima.

Table 2: Reference and alternative settings [m^{-2}], with reference gradient [T/m] and assumed-bore pole field [T]. Negative K indicates defocusing in x.

Quad	Reference K	Alternative K	Reference G	$\lvert B_{\mathrm{pole}}\rvert$
q11	-0.367525	0.060274	-4.904	0.095
q12	-1.216433	-2.126974	-16.230	0.314
q13	1.586072	1.869755	21.162	0.409
q21	-1.068373	0.954499	-14.255	0.276
q22	2.823108	-0.258649	37.667	0.729
q23	-2.490968	-1.197779	-33.236	0.643
q31	2.042678	2.817163	27.255	0.527
q32	-1.434271	-3.000000	-19.137	0.370
q33	0.351115	0.901619	4.685	0.091

For local sensitivity diagnostics we normalize rows by the optical scales and columns by an assumed common 0.01 m^{-2} tuning increment: $A_{ij} = (\partial O_i/\partial K_j)\Delta K_j/\sigma_i$. Central steps $10^{-4}, 10^{-5}, 10^{-6}$ m^{-2} converge; the six active modes and three-dimensional local null space have condition number 57.78074. Six finite small-mode perturbations remain inside bounds and agree with linear predictions within 1%. This differs from the archived mixed-unit condition number 1421.636. Neither Euclidean singular-vector orthogonality nor a local null space guarantees finite optical decoupling. Achieved commissioning correction claims are withdrawn: measurements, actuator calibration and estimation uncertainty have not been established.

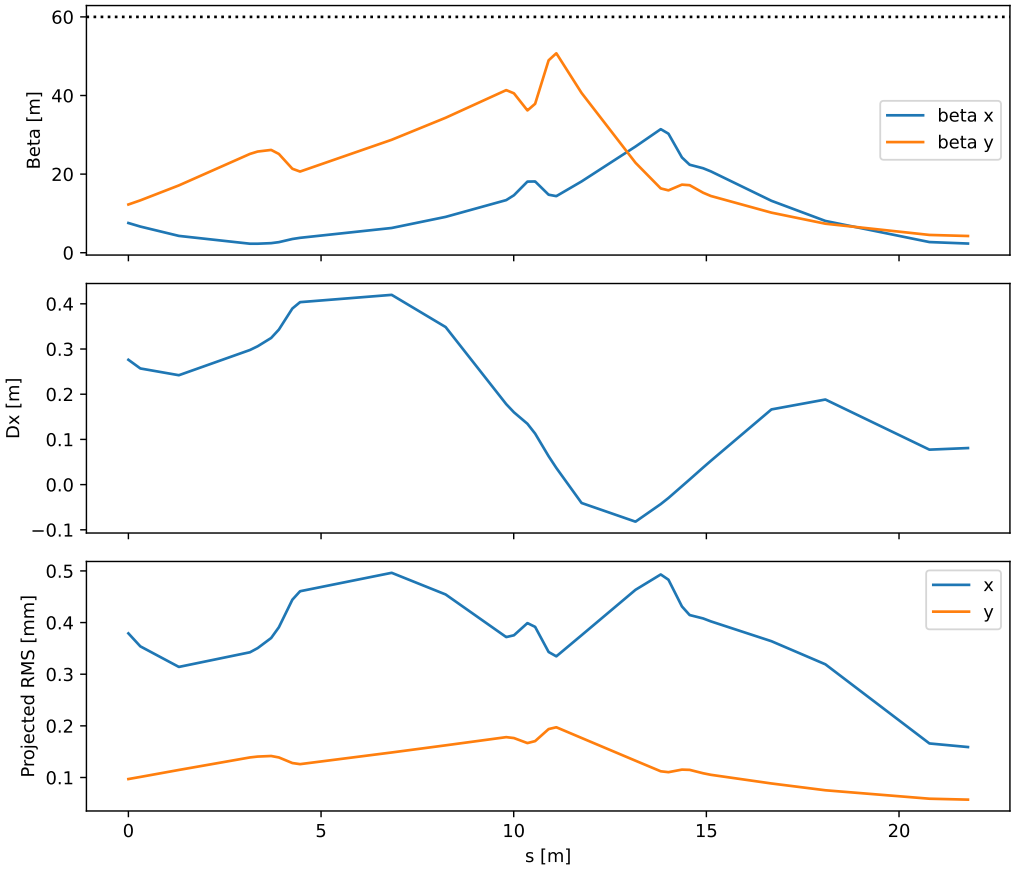


Figure 1: Reference element-boundary beta, horizontal dispersion and projected RMS sizes for the design beam. The dotted beta line is the assumed optical screening ceiling, not a vacuum-chamber edge.

2 Historical optical statistics

Reconstruction of all 550,000 historical observations confirms five optical-rule exceedances, no invalid evaluations, and compatible model/target/error inputs across four runs. The rule is $\beta_{x,\max}$ or $\beta_{y,\max} > 60.01$ m, or $\mathcal{M}_x + \mathcal{M}_y > 2.05$, where $\mathcal{M}_u = \frac{1}{2}\mathrm{Tr}(\Sigma_{u,t}^{-1}\Sigma_{u,o}) - 1$ uses unit-emittance transverse matrices. All five failures exceed vertical beta only (60.23–61.64 m); they are not physical losses.

The descriptive optical-failure estimate is $5/550000 = 9.09 \times 10^{-6}$ with exact asymmetric 95% limits $[2.95 \times 10^{-6}, 2.12 \times 10^{-5}]$; Wilson limits are $[3.8831 \times 10^{-6}, 2.1283 \times 10^{-5}]$ [1, 2]. The pooled horizontal median 0.00208629 has the order-statistic interval [0.00207403, 0.00209890], not a pooled-bootstrap interval. Reconstructed median, 95th and 99th quantile intervals match the archived analysis.

Three 50,000-sample median-prefix changes (20.830, 11.436 and 10.567)$\times 10^{-6}$ miss the 10^{-5} target; the 400,000 follow-up change 2.395×10^{-6} passes. This is median stability, not rare-tail convergence. The follow-up stopping decision is not archived as predeclared; fixed-N intervals do not certify sequential coverage. Historical one-at-a-time sensitivity is mean absolute change of $\mathcal{M}_x + \mathcal{M}_y$, not J. Seed42 ranks beta-only entrance errors (0.00890374), gradients (0.000861122) and energy (0.000106336). Near-zero alignment/roll mismatch does not certify alignment/coupling tolerances. Historical and new-model data remain separate.

3 Finite loss events and numerical validation

The new inference export contains exact block binomial intervals and independent reconstruction of all nine paired clearance intervals. Zero losses among 200 independent three-shot blocks gives one-sided 95% upper event probability $1 - 0.05^{1/200} = 0.014867$. A two-sided descriptive binomial calculation for zero losses among 1.2 million particles gives upper probability 3.0741×10^{-6}; this requires an independence assumption not justified as unconditional machine reliability. Exact-binomial attribution is Clopper and Pearson [2].

The complete source-labelled covariance, linearization about actual shot centroids, uncensored nonlinear Gaussian quadrature, integration/probe convergence and spatial meshes are available in the attachment. Quadrature certifies moments only on checked cases, not rare-tail loss probabilities. Comparison retains first-loss elements, coupled covariance and exact random seeds. Station transmission ends at m4; unknown continuous chamber and ring capture remain outside the model.

The initial shared-aperture hypothetical run is excluded and retained with exact generating sources. The corrected driver independently detaches the selected station, verified by a regression test. No numerical failure record is silently counted as physical success.

References

[1] E. B. Wilson. Probable inference, the law of succession, and statistical inference. *J. Am. Stat. Assoc.*, 22:209–212, 1927. `https://doi.org/10.1080/01621459.1927.10502953`.

[2] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. `https://doi.org/10.1093/biomet/26.4.404`.